



Lossless & Lossy Decomposition

$R(A, B, C, D, E)$ with $FDs = \{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$ state which of the following is lossless and lossy decomposition

a) $R_1(C, E)$ $R_2(A, B)$ $R_3(A, D, E)$

Sol :- Identify PK in all relations (Step #01)

$R_1(\underline{C}, \underline{E})$ $R_2(\underline{A}, \underline{B})$ $R_3(\underline{A}, \underline{D}, \underline{E})$

(Step #02) Take $R_1 \cup R_2 \cup R_3$ and check is it equal to R
so $R_1 \cup R_2 \cup R_3 = (C, E, A, B, D) = R$
so it is true

(Step #03) Identify those relations in pair that has something in common

$R_1 \& R_3$

$R_2 \& R_3$

Step #04 Take intersection of these pair. If they contain PK of any of the relation the decomposition is lossless otherwise it is lossy.

$R_1 \cap R_3 = \{E\} \rightarrow$ PK of R_3 hence True } overall it is lossless!
 $R_2 \cap R_3 = \{A\} \rightarrow$ PK of R_2 hence True }

b) $R_1(C, E)$ $R_2(A, B)$ $R_3(A, C, D)$

Sol:- $R_1(C, \underline{E})$ $R_2(\underline{A}, B)$ $R_3(A, \underline{C}, \underline{D})$

$$R_1 \cup R_2 \cup R_3 = \{C, E, A, B, D\} = R \quad \checkmark$$

$R_1 \& R_3$ $R_2 \& R_3$

$R_1 \cap R_3 = \{C\}$ as C is not the PK of any of R_1 and R_3
hence, the composition is lossy